

# Elodie Germani

## Postdoctoral Research Associate

*Medical Imaging, Artificial Intelligence, Biostatistics*

## About me

Computational imaging researcher, specialized in healthcare imaging data and biomarkers. Currently working on AI-based tools to extract robust and personalized information from medical images of organs and systems.

**Current situation :** Postdoctoral researcher at Université de Rennes (France) since September 2025.

**PhD Defense :** September 2024, at Université de Rennes

**Teaching Qualification :** Sections 27 (Computer sciences) and 61 (Signal Processing), since 2025.

**Email :** elodiegermani@gmail.com

**Website :** <https://elodiegermani.fr>

**Google Scholar :** <https://scholar.google.com/citations?user=BTAAgQEAAAAJ>

**GitHub :** <https://github.com/elodiegermani>

## Research experiences

---

### 09/2025 - Present

*Université de Rennes,  
France*

### Postdoctoral Researcher

*Medicis, UMR Inserm U1099 Laboratoire Traitement Signal et Images*

Personalized medical imaging biomarkers for women's health.

- Lesion detection in pelvic MRI for endometriosis surgical planning [8];
- Robustness metrics in prompt-based segmentation models [7];
- Clinically-guided radiomics features in female pelvic MRI [9], [22].

---

### 09/2024 - 09/2025

*Universitätsklinikum Bonn,  
Allemagne*

### Postdoctoral Researcher

*Albarqouni Lab, University Hospital Bonn*

Fairness and biases of AI-based tools for medical image analysis.

- Bias in Foundation Models : a multi-site study on mammography [19], [1];
- Diffusion models for quality transfer in pediatric ultra-low-field MRI [11].

Supervised by : Pr. Shadi Albarqouni

---

### 06/2023 - 12/2023

*Concordia et McGill  
University, Montreal,  
Canada*

### Visiting Researcher (MITACS Fellowship)

*Big Data for Neuroinformatics lab (Concordia) ORIGAMI Lab (McGill)*

LivingPark : reproduction and robustness evaluation of studies on MRI biomarkers for Parkinson's disease. Impact of methodological variations on the performance of prediction models based on resting-state fMRI biomarkers [4],[15].

Supervised by : Pr. Tristan Glatard and Pr. Jean-Baptiste Poline.

---

### 09/2021 - 09/2024

*Université de Rennes,  
France*

### Doctoral Researcher

*IRISA - UMR 6074 (Empenn and LACODAM)*

Representation learning to explore and mitigate the effect of analytical variability in functional MRI studies.

- Data reuse to improve the robustness of neuroimaging studies [3],[5];
- Low-dimensional representation of fMRI data for decoding studies [6].

Supervised by : Dr. Camille Maumet and Pr. Elisa Fromont.

---

|                                                                                         |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>01/2021 - 07/2021</b><br><i>Université de Rennes 1,<br/>France</i>                   | <b>Research Internship</b><br><i>IRISA - UMR 6074 (Équipe Empenn)</i><br>Developing a codebase reproducing fMRI pipelines used in a many analyst study (Narps Open Pipelines project). Quantification of variability across pipeline results and evolution with sample size in group-level statistics [11].<br>Supervised by : Dr. Camille Maumet. |
| <b>04/2020 - 07/2020</b><br><i>Centre d'Immunologie de<br/>Marseille-Luminy, France</i> | <b>Research Internship</b><br><i>IMAGIMM Laboratory</i><br>Development of an R/Shiny application to facilitate histopathology data analysis. The application features an interactive interface linking images to quantitative measurements (graphs) [6].<br>Supervised by : Mathieu Fallet and Dr. Hugues Lelouard.                                |

## Diploma

|                                                                      |                                                                                                                                                                                             |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2021 - 2024</b><br><i>Université de Rennes</i>                    | <b>Doctoral degree in Computer Sciences</b><br><i>"Exploring and mitigating analytical variability in functional MRI with representation learning", defense on 16/09/2024.</i>              |
| <b>2019 - 2021</b><br><i>Université de Rennes 1</i>                  | <b>Master's degree in Bioinformatics</b><br>Sp. Integrative Biology and Computer Sciences. Mention "Très Bien" (1/17).                                                                      |
| <b>2015 - 2018</b><br><i>Université de Versailles<br/>St-Quentin</i> | <b>License degree in Health Sciences</b><br>Diploma of General Training in Medical Sciences (DFGSM)<br>First year common to Health Studies (PACES) - <i>Succeeded in one year (70/699).</i> |

## Awards & Grants

- **Fondation APICIL** (10,000€, in 2026)  
Funding for starting a project on brain imaging signatures of pain in women with endometriosis.
- **L'Oréal-UNESCO Young Talents Award - For Women In Science** (15,000€, in 2024)  
Selected for my doctoral research along with 34 other PhD and postdoctoral researchers from over 800 applications. This award aims to promote the careers and contributions of young women in science. [Press Book]
- **IABM Best Poster in Medical Imaging** (500€) and **Travel Award** (400€) in 2024.
- **MITACS Global Research Award** (6,000 CAD, in 2023)  
Mobility grant for a research stay in the lab of Prof. Tristan Glatard at Concordia University, Montreal.
- **Rennes Métropole Mobility Grant** (2,400€, in 2023)  
Additional funding for a research stay in Montreal, Canada, in Prof. Jean-Baptiste Poline's lab (ORIGAMI)

## Scientific community

### Invited talks

- **Seminars at international research institutes**
    - ORIGAMI Lab, McGill University (Montreal), July 2023
    - Big Data for Neuroinformatics Lab, Concordia (Montreal), September 2023
    - Center for Medical statistics, Medical University Vienna (Austria), February 202
    - IPI, LS2N (Nantes), March 2025
    - MediCIS, LTSI (Rennes), April 2025
  - **FAIMI Virtual Workshop on Fairness of AI in Medical Imaging**  
Presentation of my work on bias and generalizability of Foundation Models for Breast Mammography classification.  
Online - 25/11/2025
  - **Franco-German Day for Young Researchers**  
Exchange and panel discussion on healthcare, numeric and artificial intelligence.  
Institut Français, Berlin (Germany) - 21/10/2025.
  - **“Women in Science: Breaking Frontiers – A Data Perspective”**  
Panel discussion on the importance of data diversity in health-related artificial intelligence.  
Institut Français, Vienna (Austria) – 11/02/2025.
- 

### Collaborative research projects

- **BRIDGE Project - Participating researcher (2025 - ongoing)**  
**The project** : development of international data governance frameworks to enable the equitable sharing of brain MRI data. **My contribution** : I accessed, preprocessed, and analyzed three resting-state fMRI datasets and documented the practical requirements for these steps; I reported the results of these analyses comparing healthy control groups with pathological groups in resting-state fMRI.
  - **Narps Open Pipelines - Project lead (2021 - 2024), Participating researcher (2024 - ongoing)**  
**The project** : The goal of the NARPS Open Pipelines project is to create a codebase reproducing the 70 pipelines from the NARPS study and to share it as an open resource for the community. **My contribution** : As the initiating developer, I created the code skeleton and set up the first pipelines; I presented this project at several hackathons. As a participant, I reproduced pipelines and took part in project events.
- 

### Scientific events and networks

- **European Fall School on Endometriosis 2025** : Selected candidate for attendance and short presentation.
  - **Femme de Tech Network** : Member of the Femme de Tech Network at the Académie des Sciences (2025-).
  - **German Postdoc Network** : Member of the working group for the annual conference (2024-2025).
  - **FLUID Workshop (AAAI 2025)** : Member of the outreach committee.
  - **Summer School on Affordable AI 2024 (AGYA)** : Involved in organization, logistics, mentoring, and tutoring.
  - **Narps Open Pipelines Hackathons** : Internal organizer (IRISA and ORIGAMI lab), external project manager at OHBM Brainhack 2022 and 2023.
  - **IDA 2022** : Member of the student committee. Organization and logistics.
- 

### Reviewing

- **Conferences** : reviewer for MIDL 2025 (Salt Lake City, USA) and 2026 (Taipei, Taiwan), MICCAI 2024 (Marrakech, Morocco). Area Chair (meta-reviewer) for MICCAI 2025 (Daejeon, South Korea) and MICCAI 2026 (Strasbourg, France).
- **Journals** : Scientific Report (4), npj Digital Medicine (4), Imaging Neurosciences (1), Nature Biomedical Engineering (2), Neurolmage: Clinical (1).

## Supervision

- **Albane Blaes – Faculté de Médecine, Université de Rennes (May 2026 - Ongoing)**  
Bibliographic review of AI-based methods in MRI and ultrasound for preoperative planning of deep endometriosis.
- **Anne-Sophie Marc – Faculté de Médecine, Université de Rennes (September 2025 - Ongoing)**  
Medicine thesis on pelvic MRI radiomics biomarkers to predict surgical outcomes and quality of life in endometriosis. Guidance and co-supervision (25%) with Pre. Krystel Nyangoh-Timoh.
- **Eliot Leguy – Polytech Nantes (October 2025 - Ongoing)**  
Master's project and research internship on "Extracting robust radiomic biomarkers from female pelvic MRI". Supervision (100%) and co-guidance with Dr. John SH Baxter.
- **Ilayda Selin Türk – TU Munich (November 2024 – March 2025)**  
Master's Thesis – "Investigating Bias in AI Algorithms for Breast Cancer Detection from Mammography Imaging: A Focus on Generalization to Unseen Populations" Co-supervised (50%) with Pr. Shadi Albarqouni.
- **Malek Al Abed – TU Munich (October 2024 – July 2025)**  
Master's Thesis – "Deep Learning for Low-field Brain Image Quality Transfer" Co-supervised (25%) with Prof. Shadi Albarqouni. Additional supervision : Pr. Hemmen Sabir, Dre. Anne Groteklaes, and Dre. Sebiha Demir.
- **Thibault Chanus – ENS Rennes (October 2023 – January 2024)**  
Master 1 Project – Diffusion models for style transfer tasks. Co-supervised (50%) with Pre. Elisa Fromont.
- **Maiwenn Fleig – Düsseldorf University (April 2023 – July 2023)**  
Engineering Project – Use of diffusion models for functional MRI data. Co-supervised (50%) with Pre. Elisa Fromont.

## Teaching

During my doctoral and postdoctoral experiences, I taught various courses in **data science, machine learning, and computer science**. In total, I have taught more than **250 hours-equivalent TD (HETD)**. The audience was diverse and interdisciplinary, reflecting both my academic background and research interests. Full description in Annex 1 .

### Data science

- **Mathematics** in L1 Biology (UR) : introduction to Python programming, data analysis, probability laws ;
- **Data science** in L2 Computer sciences (UR) : statistical testing, data modeling, and time series analysis in R ;
- **Basics of Data Analysis** in M1 Informatics for Business (UR) : introduction to data management in Python ;
- **Big Data Analysis** in M2 Signals and Images in Biology and Medicine (UR) : biases, concept of false positive/negative findings, processing pipelines, and statistical testing.

### Machine learning and applied AI

- **Machine learning** in M2 Computer sciences (UR) : Introduction to machine learning in Python ;
- **AGYA Summer School on Affordable AI** (UKB) : Introduction to machine learning in Python ;
- **AiR Seminars "Artificial Intelligence in Radiology: Applications and Research"** for medicine students (UKB) : Teaching the role of AI tools in medical imaging : classification, detection, prediction, segmentation ;
- **Artificial Intelligence for Medical Diagnosis & Prediction** for medicine students (UKB) : Creation of hands-on tutorials (notebooks in Python) to teach medical data analysis with artificial intelligence.

### Computer science

- **Computer systems** in L1 Computer science (UR) : Python programming to model computer components ;
- **Database management** in L2 and M1 Computer Science (UR) : Project on the comparison of different database management systems (PostgreSQL, MonetDB, etc.) ;
- **Applied programming** in L3 Electronics (UR) : initiation to programming with Python, C, and Matlab.

## Scientific outreach

### Public

- **Pint of Science** : Event manager for the 2024 edition, under the theme “*Our Body*”.
- **Brain Awareness Week** : Designed and hosted a quiz : “*Research Careers in Neuroimaging*”. Participated in the 2022 and 2024 editions (project lead in 2024). [Materials]  
Invited presentation for the 2026 edition : “Chronic Pain Signatures in Women’s Brains”. [Event][Materials]
- **Open Days MediCIS** : Member of the organizing committee for the laboratory’s open days (February 10–11, 2026), responsible for logistics and demonstrations.
- **TISSAGE** : Training in science communication offered by the Doctoral School as part of the TISSAGE project. Led a quiz on the topic “*Artificial Intelligence vs. Human Intelligence*”. [Materials]
- **Media** : SQYMag [Video], TVRennes [Video], Ouest France [Article], and others [Press Book].

### Scholars

- **Journées Filles, Maths et Informatique : une équation lumineuse** : Opening conference (invited) on artificial intelligence for medical imaging at the 2025 edition at the Université de Rennes.
- **L Codent, L Créent** : Creative Python programming workshops for middle school girls to promote and demystify computer science. Participated from 2021 to 2023. Presentation at the Femmes et Sciences Colloquium in 2022. [Materials]
- **Computing without computer** : Workshops about computer science serious games. [Website]
- **CHICHE** : Presentation of the research academic path and of my research topic in middle and high schools (College Malherbe de Chateaubourg, Lycée Loth de Pontivy).
- “**J’peux pas j’ai informatique**” : Workshops about computer science and women for middle-/high-school teachers.

## Other experiences

|                                    |                                                                                                                                                                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2025</b><br><i>SoluVitrages</i> | <b>Web Development</b><br><br>Development of a web tool for managing appointments, inventory, and revenue. Python programming with Flask, connected to an SQL database.                                                                                             |
| <b>2019</b><br><i>SCC France</i>   | <b>IT Support Technician (Level 1)</b><br><br>Assignment at the Ministry of Armed Forces (Paris, France). Scheduling and appointment management, computer replacement, technical support, and inventory management.                                                 |
| <b>2017–2018</b><br><i>i-Share</i> | <b>Communication Assistant</b><br><br>i-Share is a cohort study on student health ( <a href="https://www.i-share.fr/">https://www.i-share.fr/</a> ). Tasks included promoting the study to students : setting up booths, giving presentations at forums, open days. |

## Technical skills

|                    |                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python : Expert.<br><i>Deep learning (PyTorch, scikit-learn), Data (NumPy, pandas), Visualization (matplotlib, seaborn).</i><br>R : Intermediate<br>HTML/CSS : Intermediate<br>Java : Beginner<br>Matlab : Beginner |
| <b>Software</b>    | Neuroimaging (FSL, SPM) : Advanced                                                                                                                                                                                  |

---

**Other**

$\LaTeX$  : Advanced

Git : Advanced

Docker/Apptainer : Intermediate

---

## Publications

Here are my publications published or accepted in peer-reviewed international journals and conferences. In artificial intelligence and medical imaging, the first authors are those who led the work, and the last authors are the supervisors. Journal rankings are based on Scimago, and conference rankings come from ICORE.

\* indicates equal contribution among co-authors.

## Journal Papers

**[1] When Multi-Institutional Dataset Aggregation Masks Shortcut-like Behavior in Foundation Models for Medical Imaging**

Germani, E., Zeineddine, F., Mourad, C., Albarqouni, S.

*Medical Image Analysis (Special Issue MICCAI 2025)* 114 :104258. DOI : [10.1016/j.media.2026.104258](https://doi.org/10.1016/j.media.2026.104258).

*Invitation to submit a journal extension of International Conference Paper [12]*. Medical Image Analysis is one of the leading journals in medical image analysis and was Q1 in Health Informatics in 2025.

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code]

**[2] The HCP multi-pipeline dataset : an opportunity to investigate analytical variability in fMRI data analysis**

Germani, E., Fromont, E., Maurel, P., Maumet, C.

*Sci Data* 12, 940. DOI : [10.1038/s41597-025-05247-7](https://doi.org/10.1038/s41597-025-05247-7).

Dataset available in [Public nEUro](#). *Scientific Data* was Q1 in Computer Science Applications in 2022.

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code]

**[3] On the validity of fMRI studies with subject-level data processed through different pipelines**

Germani, E.\*, Rolland, X.\*, Maurel, P., Maumet, C.

*Imaging Neuroscience*; 3. DOI : [10.1162/imag\\_a\\_00522](https://doi.org/10.1162/imag_a_00522).

*Imaging Neuroscience* is a young journal created after resignation of NeuroImage Editorial Board (Q1 in Neuroscience in 2023) following an increase in APC.

Contributions : Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code]

**[4] Predicting Parkinson's disease trajectory using clinical and functional MRI features : a reproduction and replication study**

Germani, E., Baghwat, N., Dugré, M., Gau, R., Montillo, A. A., Nguyen K. P., Sokolowski A., Sharp, M., Poline JB., Glatard, T.

*PLOS ONE* 20(2) : e0317566. DOI : [10.1371/journal.pone.0317566](https://doi.org/10.1371/journal.pone.0317566).

In 2023, PLOS ONE was Q1 in "Multidisciplinary".

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code]

**[5] On the benefits of self-taught learning for brain decoding**

Germani, E., Fromont, E.\*, Maumet, C.\*

*GigaScience*, 2023, Vol. 12, pp.1-17. DOI : [10.1093/gigascience/giad029](https://doi.org/10.1093/gigascience/giad029).

In 2023, *GigaScience* was Q1 in "Computer Science Applications and Health Informatics".

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code] [Supplementary]

**[6] SAPHIR : a Shiny application to analyze tissue section images**

Germani, E., Lelouard, H., Fallet, M.

*F1000Research*, 2020, 9, pp.1276. DOI : [10.12688/f1000research.27062.2](https://doi.org/10.12688/f1000research.27062.2).

In 2020, *F1000Research* was Q1 in "Medicine (miscellaneous)". *F1000Research* is a journal with post-review, the

manuscript has been approved after revision by 2 reviewers. The paper is in the category “Software Tool Article”.  
Contributions : Conceptualization, Methodology, Programming, Writing.  
[Article] [HAL] [Code]

## International Conference Papers

**[7] Disentangling Prompt Dependence to Evaluate Segmentation Reliability in Gynecological MRI**

Germani, E., Nyangoh Timoh K., Jannin P., Baxter, J.S.H.

*International Conference on Medical Image Computing and Computer-Assisted Intervention, MICCAI 2026.*

Accepted with poster and oral presentation (acceptance rate  $\sim 28\%$ , with less than 10% selected for oral).

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Preprint] [HAL]

**[8] Comparing Problem Formulations for Endometriosis Lesion Detection on T2-weighted Pelvic MRI**

Leguy, E., Nyangoh Timoh, K., Jannin, P., Baxter, J.S.H., Germani, E.

*MICCAI Workshop on Computer-Assisted Pelvic Imaging (CAPI) and Women's Health (WOMEN) 2026.*

Accepted with poster and oral presentation.

Contributions : Conceptualization, Data Curation, Writing - Review & Editing, Supervision.

[HAL]

**[9] Multi-scale radiomics in pelvic MRI for endometriosis subtyping : highlighting data heterogeneity constraints**

Leguy, E., Mallet, C., Normand, N., Germani, E.

*MICCAI Workshop on Computer-Assisted Pelvic Imaging (CAPI) and Women's Health (WOMEN) 2026.*

Accepted with poster presentation.

Contributions : Conceptualization, Data Curation, Writing - Review & Editing, Supervision.

[Preprint] [HAL]

**[10] Foundational values for foundation models**

Baxter, J.S.H., Germani, E.

*MICCAI Workshop on Fairness, Regulation, and Ethics (FAIMI-BRIDGE-EPIMI 2026).*

Accepted with poster and oral presentation.

Contributions : Conceptualization, Writing - Review & Editing.

[Preprint]

**[11] MRIQT : Physics-Aware Diffusion Models for Image Quality Transfer in Neonatal Ultra-Low-Field MRI**

Al Abed, M., Demiha, S., Groteklaes, A., Germani, E., Hemmen, S., Albarqouni, S.

*International Symposium on Biomedical Imaging (ISBI) 2026.* Presentation : Oral.

Contributions : Data curation, Supervision, Formal analysis.

[Article] [HAL]

**[12] Bias and Generalizability of Foundation Models across datasets for Breast Mammography Classification**

Germani, E., Selin-Türk, I., Zeineddine, F., Mourad, C., Albarqouni, S.

*International Conference on Medical Image Computing and Computer-Assisted Intervention, MICCAI 2025.*

DOI : 10.1007/978-3-032-05185-1\_3

Early-acceptance (9% of submission). Spotlight and poster presentation, short-listed for Best Paper Award.

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL]

**[13] Mitigating analytical variability in fMRI results with style transfer**

Germani, E., Fromont, E.\*, Maumet, C.\*

*Conference in Medical Imaging with Deep Learning - MIDL 2025.*



Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[HAL] [Code]

**[14] Uncovering communities of pipelines in the task-fMRI analytical space**

Germani, E., Fromont, E.\*, Maumet, C.\*

2024 IEEE International Conference on Image Processing (ICIP), Abu Dhabi, United Arab Emirates, 2024, pp. 3044-3050. DOI : 10.1109/ICIP51287.2024.10647701.

Conference specialized in image processing, ranked A-B. Presentation : Oral.

Contributions : Conceptualization, Methodology, Programming, Analysis, Writing.

[Article] [HAL] [Code] [Slides]

## International Conference Abstracts

**[15] Exploring variability patterns in the task-fMRI analytical space**

Germani, E., Fromont, E.\*, Maumet, C.\*

OHBM 2023 - 29th Annual Meeting of the Organization for Human Brain Mapping. Jul 2023, Montreal, Canada. Presentation : Poster.

[Poster]

**[16] FMRI data analysis : How does analytical variability vary with sample size ?**

Germani, E., Maumet, C.

OHBM 2022 - 28th Annual Meeting of the Organization for Human Brain Mapping. Jun 2022, Glasgow, United Kingdom. pp.1-5. Presentation : Poster.

[Poster] [Code]

## National Conferences Abstracts

**[16] Bridging the gap toward automated analysis of female pelvic MRI**

Germani, E., Baxter, J., Jannin, P., Nyangoh-Timoh, K.

IABM 2026 - 6ème édition du Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale. Mar 2026, Lyon, France.

[Poster]

**[17] Prédire l'évolution de la maladie de Parkinson à l'aide de données cliniques et d'IRM fonctionnelles : reproduction et robustesse d'une étude**

Germani, E., Baghwat, N., Dugré, M., Gau, R., Sokolowski, A., Sharp, M., Poline, JB.\*, Glatard, T.\*

IABM 2024 - 2ème édition du Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale. Mar 2024, Grenoble, France. Presentation : Poster. **Best poster in Medical Imaging award.**

[Poster]

**[18] Representation learning for more reproducible fMRI data analyses**

Germani, E., Fromont, E.\*, Maumet, C.\*

IABM 2023 - Colloque Français d'Intelligence Artificielle en Imagerie Biomédical. Mar 2023, Paris, France. Presentation : Poster.

[Poster]

## Other valorisations

**[19] Proceedings of the OHBM Hackathon 2023**

Yang, Y.-F., Heinsfeld, A. S., Gondová, A., Vieira, B. H., Wang, Q., Mansour, S. L., Li, X., Haugen, B., Bandettini, P., Bryant, A., Chakravarthy, M. M., Clarke, N., Clénet, B., Clucas, J. C., Dugré, M., Earl, E., Germani, E., Goodale, S. E., Gulban, O. F., Huber, R., Jeganathan, J., Joulot, M., Kai, J., Sitek, K. R., Koiso, K., Korbmacher, M., Maullin-Sapey, T., Maumet, C., McPherson, B. C., Meisler, S. L., Mentch, J., Miedema, M., Moia, S., Nikolaidis,

A., Taylor, P. A., Picard, M.-E., Pizzuti, A., Provins, C., Savary, E., Steinkamp, S. R., Taschler, B., Van Dyken, P. C., White, T., Yu, J.-C., Zou, Y., Haak, K. V.

Aperture Neuro, 2025, Vol. 5. DOI : 0.52294/001c.145051.

Summary of projects presented at OHBM Brainhack 2023. Presentation of the Narps Open Pipelines project.

[Article]

**[20] Proceedings of the OHBM Brainhack 2022**

Moia, S., Wang, H.-T., Heinsfeld, A. S., Jarecka, D., Yang, Y. F., Heunis, S., Svanera, M., De Leener, B., Gondova, A., Kim, S., Basavaraj, A., Bayer, J. M. M., Bayrak, R. G., Bazin, P.-L., Bilgin, I. P., Bollmann, S., Borek, D., Borghesani, V., Cao, T., Chen, G., De La Vega, A., Dresbach, S., Ehse, P., Ernsting, J., Esteves, I., Ferrante, O., Garner, K. G., Gau, R., Germani, E., Ghafari, T., Ghosh, S. S., Goodale, S. E., Gould Van Praag, C. D., Guay, S., Gulban O. F., Halchenko, Y. O., Hanke, M., Herholz, P., Heuer, K., Hoffstaedter, F., Huang, R., Huber, R., Jensen, O., Keeratimahat, K., Kosciessa, J. Q., Lukic, S., Magielse, N., Markiewicz, C. J., Martin, C. G., Maumet, C., Menacher, A., Mentch, J., Monch, C., More, S., Muller, L., Muller-Rodriguez, Leonardo, Nastase, Samuel A., Nicolaisen-Sobesky, E., Nielson, D. M., Nolan, C. R., Paugam, F., Pinheiro-Chagas, P., Pinho, A. L., Pizzuti, A., Poldrack, B., Poser, B. A., Rocca, R., Sanz-Robinson, J., Sarink, K., Sitek, K. R., Sychala, N., Stirnberg, R., Szczepanik, M., Torabi, M., Toro, R., Urchs, S. G. W., Valk, S. L., Wagner, A. S., Waite, L. K., Waite, A. Q., Waller, L., Wishard, T. J., Wu, J., Zhou, Y., Bijsterbosch, J. D.

Aperture Neuro, 2024, Vol. 4. DOI : 10.52294/001c.92760.

Summary of projects presented at OHBM Brainhack 2022. Presentation of the Narps Open Pipelines project.

[Article]

**[21] HCP multi-pipelines [dataset]**

Germani, E., Fromont, E., Maurel, P.\*, Maumet, C.\*

PublicNeuro Datasets. DOI : 10.70883/gtkk1541.

Multi-pipeline dataset to facilitate the exploration of analytical variability across different contexts. The dataset is available on the Public-nEUro platform and can be accessed upon signing a Data User Agreement.

[Data]

## Préprints soumis, en review, en révision

**[22] Patient-level Aggregation of Radiomic Features in Multifocal Disease : a Study on Preoperative MRI Biomarkers of Surgical Response to Deep Infiltrating Endometriosis Lesion Resection**

Germani, E., Marc, A.S., Jannin, P., Baxter, J.S.H., Nyangoh-Timoh K.

*Submitted to SPIE Medical Imaging 2026, special track "Image-Guided and Robotic Interventions".*

## Annex 1 : Courses and teaching duties

| Year                              | Institution                                           | Audience                                        | Course                                                                         | Teaching Load (HETD) | Type                             | Tasks / Materials                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------|----------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021<br>2022                      | Faculty of Mathematics<br>University of Rennes 1      | L1 Biology                                      | Mathematics for Biology                                                        | 45h                  | Lab (18h)<br>Tutorial (27h)      | Intro to programming and data analysis in Python.<br><br>Statistics and probability tutorials (conditional probability, discrete and continuous distributions).<br><br><i>Session preparation (exercises, slides).<br/>Grading assignments.</i>                                                                                                                                                 |
| 2021<br>2023                      | Fondation Rennes 1                                    |                                                 | L Codent, L Créent                                                             | 20h                  | Lab (2×10h)                      | Creative programming in Python with middle school girls to promote computer science.                                                                                                                                                                                                                                                                                                            |
| 2022<br>2023<br>+<br>2025<br>2026 | Faculty of Computer Science<br>University of Rennes 1 | L1 Computer Science                             | Computer Systems Principles                                                    | 48h                  | Lab (2×20h)<br>Tutorial (2×4h)   | Introductory Python programming labs. Critical understanding and analysis : components and architectures of computer systems.<br><br><i>Project supervision : literature review on natural language processing.</i>                                                                                                                                                                             |
| 2022<br>2023                      | Faculty of Computer Science<br>University of Rennes 1 | L1 Computer Science                             | Professional Project & Communication                                           | 8h                   | Lab (8h)                         | Introduction to word processing with OpenOffice.                                                                                                                                                                                                                                                                                                                                                |
| 2022<br>2024                      | Faculty of Computer Science<br>University of Rennes   | M2 Computer Science, Software Engineering track | Machine Learning                                                               | 20h                  | Lab (2×10h)                      | Introduction to machine learning in Python (scikit-learn, Keras). Critical understanding : hyperparameter selection, validation, performance metrics.<br><br><i>Design of teaching materials (exercises, datasets).<br/>Project supervision : software for handwritten letter and digit recognition.</i>                                                                                        |
| 2023<br>2024<br>+<br>2025<br>2026 | Faculty of Computer Science<br>University of Rennes   | L2 Computer Science                             | Data Science                                                                   | 48h                  | Lab (2×12h)<br>Tutorials (2×12h) | Introduction to programming in R for data analysis. Critical understanding : modeling and statistical testing, data visualization, time series analysis.<br><br><i>Session preparation. Grading assignments.</i>                                                                                                                                                                                |
| 2023<br>2024                      | Faculty of Medicine<br>University of Rennes           | M2 Signals and Imaging in Biology and Medicine  | Big and Complex Data Analysis                                                  | 3h                   | Lecture (3h)                     | Teaching of processing pipeline concepts, analytical variability, and publication practices.<br><br><i>Adaptation of lecture materials and exam. Grading assignments.</i>                                                                                                                                                                                                                       |
| 2024                              | Arab-German Young Academy of Sciences and Humanities  | Master's and PhD students (interdisciplinary)   | Summer School AGYA Summer School on Affordable AI                              | 10h                  | Lab (10h)                        | Introduction to machine learning in Python. Critical understanding : hyperparameter selection, evaluation, resource-constrained scenarios.<br><br><i>Preparation and delivery of sessions.</i>                                                                                                                                                                                                  |
| 2024<br>2025                      | University Hospital Bonn                              | Medical students                                | AiR Seminars "Artificial Intelligence in Radiology: Applications and Research" | 12h                  | Lecture (12h)                    | Teaching the role of AI tools in medical imaging : classification, detection, prediction, segmentation. Critical understanding : evaluation of ethical and regulatory considerations in deploying AI in healthcare – bias, uncertainty.<br><br><i>Preparation of course materials, logistics, support to instructors. Assessments : presentation and critical analysis of a research paper.</i> |
| 2024<br>2025                      | University Hospital Bonn                              | Medical students                                | Artificial Intelligence for Medical Diagnosis & Prediction                     | 16h                  | Tutorials (16h)                  | Creation of hands-on tutorials (notebooks in Python) to teach medical data analysis with artificial intelligence : data preparation and preprocessing, visualization, machine learning fundamentals, metrics and evaluation, deep learning, multimodal integration and foundation models.                                                                                                       |

|              |                         |                           |                                    |     |                                                          |                                                                                                                                     |
|--------------|-------------------------|---------------------------|------------------------------------|-----|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2025<br>2026 | University of<br>Rennes | M1 Data<br>Science        | Basics of Data<br>Analysis         | 30h | Lecture<br>(7.5h)<br>Tutorials<br>(7.5h)<br>Lab<br>(15h) | Introduction to data science and Python program-<br>ming : basics, numpy, pandas.                                                   |
| 2025<br>2026 | University of<br>Rennes | M1<br>Computer<br>Science | Advanced<br>Database<br>Management | 45h | Lab<br>(45h)                                             | Project supervision : analyzing the differences bet-<br>ween two SGBD in terms of indexing, storage, re-<br>quests, or joins.       |
| 2025<br>2026 | University of<br>Rennes | L3<br>Electronics         | Applied<br>Programming             | 15h | Lab<br>(12h)<br>Tutorials<br>(3h)                        | Introduction to Python, Matlab and C program-<br>ming languages.                                                                    |
| 2025<br>2026 | University of<br>Rennes | L2<br>Computer<br>Science | Databases                          | 36h | Lab<br>(18h)<br>Tutorials<br>(18h)                       | Introduction to the concept of databases (mo-<br>delling, normalization, etc.). Practical applications<br>with lab sessions on SQL. |